MILITARY SPECIFICATION SHEET

ELECTRON TUBE, KLYSTRON

TYPE 2K48

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Reflex, external cavity

ABSOLUTE RATINGS:

Parameter:	Ef	Cavity voltage	Ik	Er	T	Alt
Unit:	V	Vdc	mAdc	Vdc	°C	ft
Maximum:	6.6	1,500	15	-400	120	50,000
Minimum:	6.0	---	---	-18	---	---

PHYSICAL CHARACTERISTICS:

See figure 1
(D) Mounting position: Any

TEST CONDITIONS:

Parameter:	Ef	Cavity voltage	Er
Unit:	V	Vdc	Vdc
	6.3	1,250	Adjust

GENERAL:

Qualification - Required

(D) denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Qualification inspection						
(D) 1031	Nonoperation vibration	2	10 G	--	---	---	---
	Quality conformance inspection, part 1	8					
(D) 4289	Heater current	-		If	0.45	0.70	A
4229	Total reflector current	5	Er = -400 Vdc; t = 30 seconds	Ir	---	10	µA
---	Voltage breakdown	3		---	---	---	---
1256	Electrode current (cathode)	-	Er = -350 Vdc	Ik	8	15	mAdc
4250	Power output (1)	4	Cavity I: Er = -20 to -50 Vdc; F = 4,300 to 4,450 MHz	Po	5	---	mW
			Cavity II: Er = -65 to -110 Vdc; F = 6,900 to 7,200 MHz	Po	5	---	mW
4250	Power output (2)	4	Cavity II: Er = -175 to -235 Vdc; F = 6,900 to 7,200 MHz	Po	20	---	mW
4250	Power output (3)	4	Cavity III: Er = -240 to -300 Vdc; F = 10,550 to 10,850 MHz	Po	20	---	mW
(D) 4214	Cathode emission	7	Ef = 5.7 V	ΔPo/Po	---	15	%
	Quality conformance inspection, part 2						
1031	Low-frequency vibration	2	2.5 G	---	---	---	---
1036	Bump	1	Angle = 25°	---	---	---	---
---	High-altitude voltage breakdown	6		ΔIk	---	5	µA
	Quality conformance inspection, part 3						
---	Life test	-	Group C	t	250	---	hrs
---	Life-test end point:						
4250	Power output (3)	4		Po	10	---	mW

NOTES:

1. The tube shall be suspended by a yoke clamped to the cavity center conductor centered at a point 2-3/8 (60.33 mm) inches from the end of the center conductor. The weight of the yoke shall be 1/4 pound (113.4g) and the portion of the yoke against which the hammer strikes shall have a spherical radius of 1 (25.4 mm) inch.
2. The tube shall be within the limits of all power output tests after vibration.
3. A 25,000-ohm resistor shall be placed in series with the repeller lead and 4,000 peak ac volts shall be applied between the cavity center conductor and the repeller lead. No breakdown as indicated on a high-resistance ac voltmeter in parallel with the tube shall occur.

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NOTES: -Continued

4. Each tube shall be tested to assure satisfactory operation under the following conditions:

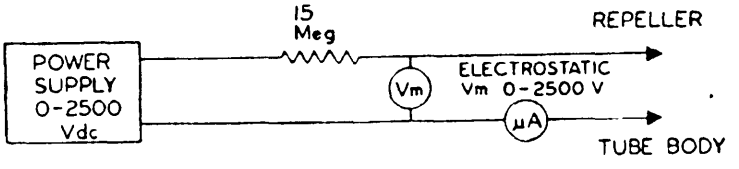
Cavity I, II, and III, in accordance with Drawing 181-JAN.

The power output shall maximize between the specified repeller voltage limits and shall be above the limits specified.

The tube shall deliver power over a repeller voltage range of ±10 V. The tube shall be assumed satisfactory without a test at ±10 V from a maximum power repeller voltage between -30 and 0 V.

5. A 15-megohm resistor shall be placed in series with the repeller lead.

6. Apply 2,000 Vdc through a 15-megohm protective resistor in series with a 50 µA full scale meter shunted by a momentary push button switch, as shown in circuit diagram below.



Current shall be read with tube at standard atmospheric pressure. The same measurement shall then be made at a pressure of 81.28 mm of mercury, care being taken not to depress the shorting button across the microammeter if any initial deflection exists. The increase in current shall not exceed 5 µA over that read at atmospheric pressure, 1 minute after application of full voltage.

7. Power output (1), test cavity I, limits shall be used.

(D) 8. The AQL for the combined defectives in quality conformance inspection, part 1, shall be 1.0, inspection level II.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 80
DSA - ES

User activities:
Army - MU
Navy - AS, OS, MC, CG, SH
Air Force - 11

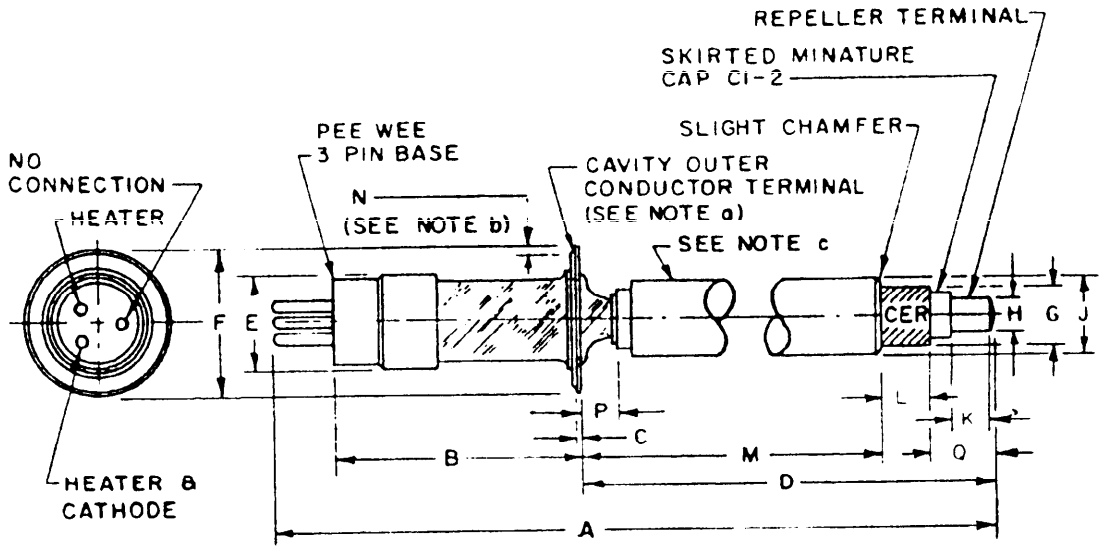
Preparing activity:
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Agent:
DSA - ES

(Project 5960-2969)

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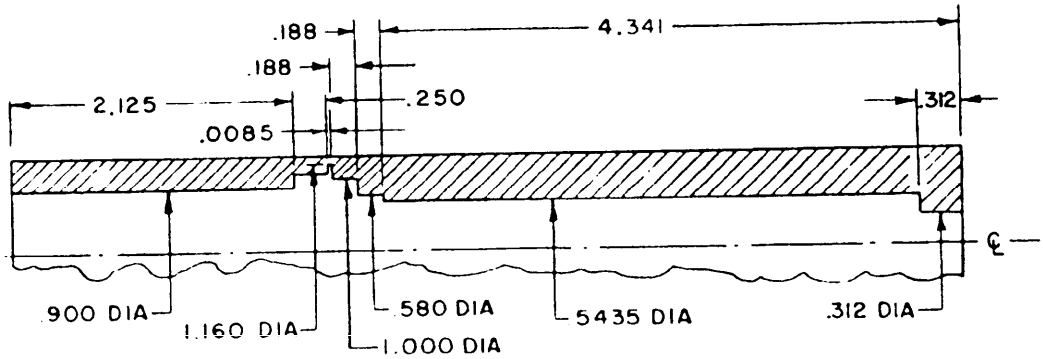


Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 1 (See note 8)		
C	.005 (.13)	.008 (.20)
F	1.125 (28.58)	1.145 (29.08)
J	.538 (13.67)	.543 (13.79)
N	.065 (1.65)	
Quality conformance inspection, part 2		
A		7.38 (187.45)
B	1.56 (39.62)	1.94 (49.28)
D	4.88 (123.95)	5.00 (127.00)
M	4.06 (103.12)	4.16 (105.66)
P	.25 (6.35)	.34 (8.64)
Quality conformance inspection, part 3		
E		.725 (18.42)
H	.245 (6.22)	.255 (6.48)
Reference dimensions		
G	.450 (11.43)	
K	.281 (7.14)	
L	.380 (9.65)	
Q	.406 (10.31)	

NOTES:
a. Shall be gold plated min 10 MSI or silver plated 15 MSI external surfaces.
b. Contact portion of disc measured radially which must be free from splits or tears, must be smooth and flat.
c. Plate cavity center conductor gold 10 MSI or silver 15 MSI.
d. The tube shall be capable of being inserted in concentric cylinder (see figure 2) without binding (equivalent gaging method may be used).

FIGURE 1. Outline drawing of electron tube type 2K48.

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($\frac{1}{4}$ SECTION OF TEST CYLINDER)

INCHES	MM
.0085	.22
.188	4.78
.250	6.35
.312	7.92
.5435	13.80
.580	14.73
.900	22.86
1.0000	25.40
1.160	29.46
2.125	53.98
4.341	110.26

FIGURE 2. Test cylinder.

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   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

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